Cutting Machine



Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Cutting machine

What it is used for

A cutting machine cuts material such as metal, wood, plastic, fabric, or other workpieces using a blade, disc, or other cutting tool. Select the correct machine and cutting accessory for the material, secure the workpiece, and follow the approved operating procedure to obtain a clean cut safely.

Main parts

- Machine frame and base
- Cutting blade, disc, or tool
- Motor and drive assembly
- Work table, vice, clamp, or material support
- Operating switch and speed or feed controls
- Protective guard, handle, and emergency stop where fitted

Before you start

1. Identify the material, required cut dimensions, and correct machine type for the job.
2. Inspect the blade or disc for cracks, wear, damage, and correct speed rating; confirm compatibility with the machine.
3. Check the guard, handle, switch, power cable, work support, and emergency stop where fitted.
4. Secure the workpiece using the correct vice, clamp, or support and confirm that the cut-off section can move without binding.
5. Remove flammable materials from the cutting area and ensure adequate ventilation and suitable spark control where required.
6. Wear task-appropriate eye and face protection, hearing protection, suitable footwear, and other required PPE; secure hair and loose clothing.

How to use

1. Switch off and isolate power before installing or changing the cutting tool or making adjustments.
2. Fit the correct blade or disc according to the manufacturer's instructions and confirm that all retaining parts are secure.
3. Adjust the guard and cutting depth or angle as required for the job without removing protective devices.
4. Position and clamp the workpiece so it cannot shift, rotate, or pinch the cutting tool during the cut.
5. Check that the cutting path is clear of clamps, supports, cables, and other obstructions.

6. Stand in a stable position with a firm grip and keep hands outside the cutting path.
7. Start the machine using the approved control and allow the tool to reach its specified operating speed before engaging the material, where applicable.
8. Feed the cutting tool into the material steadily without forcing it or twisting the blade or disc.
9. Maintain control of the machine and observe for abnormal vibration, overheating, excessive sparks, or unusual noise.
10. Stop the machine immediately if the tool binds, the workpiece shifts, the guard fails, or abnormal operation occurs.
11. Wait until the blade or disc stops completely before setting the machine down, removing offcuts, or making adjustments.
12. Switch off and isolate power before clearing jammed material or replacing a damaged cutting tool.
13. After the cut, inspect the workpiece and remove sharp burrs using an appropriate safe method.
14. Clean debris from the work area after the machine has stopped and store the tool safely.

⚠ Safety

- Only trained operators should use the cutting machine and its accessories.
- Never operate the machine with a missing, damaged, or incorrectly positioned guard.
- Keep hands, hair, clothing, and body parts away from the cutting path and rotating components.
- Never hold small workpieces by hand when they can be secured with a clamp or vice.
- Use only cutting tools rated for the machine speed and intended material; never use a cracked or damaged blade or disc.
- Do not force the tool, stand in the direct line of a potentially failing disc, or leave the machine running unattended.
- Disconnect power before changing blades, clearing jams, or performing maintenance; follow lockout procedures where applicable.
- Control sparks and dust, use appropriate respiratory protection when required, and keep suitable fire precautions in place.

Care & storage

- Clean dust, chips, and offcuts from the machine and work area after shutdown.
- Inspect the cutting tool, guard, power cable, clamps, and work supports before each use.
- Replace worn or damaged cutting tools with the correct approved type.
- Check fasteners, alignment, and lubrication points according to the manufacturer's maintenance schedule.
- Arrange servicing if the motor overheats, vibration increases, guards malfunction, or cutting accuracy deteriorates.

Common mistakes to avoid

- Using the wrong blade or disc for the material or machine speed.
- Failing to clamp the workpiece securely before cutting.
- Forcing the tool or twisting the blade during the cut.

- Removing offcuts or adjusting the machine before the cutting tool stops completely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.